

MINIMAX RECOVERY AND DECISION STABILITY FOR TREE-BASED EVIDENCE ACQUISITION

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Abstract. We study the recovery of shared acquisition costs from individual and pairwise measurements, motivated by sequential diagnosis with reusable setup resources. For nonnegative costs on a rooted tree, a spanning-tree formula recovers every set cost. With coordinatewise measurement error at most ε , the cost of $m \geq 2$ sources has minimax recovery radius $(2m - 3)\varepsilon$. A matching two-point construction establishes optimality without a separation assumption on branching depths. We give a feasible hierarchical repair and computable cost intervals for unequal measurement tolerances. These bounds yield stability guarantees for Bayesian decision values and policies, together with a certificate for a robust initial action. We also characterize universal decision equivalence and exhibit failure of pairwise recovery for general shared resources. Exact calculations and 1,920 synthetic perturbation cases illustrate the results.

Keywords: optimal recovery; rooted tree; shared acquisition cost; sequential Bayesian decision; decision stability

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1. INTRODUCTION

A sequential diagnostic procedure acquires information from specialized sources, updates its belief about a fault, and chooses a final diagnosis. Graph representations already support multi-agent scheduling in manufacturing; for example, Qin and Lu [21] combine knowledge graphs with multi-agent reinforcement learning. The present problem concerns the cost of accessing evidence when several sources share communication or session-activation resources.

Charging every query its full standalone access cost is appropriate when all communication expenses recur independently. If part of the expense is incurred once per investigation, however, later queries may reuse an activated route. A useful theoretical question is whether individual and pairwise cost measurements determine the cost of every larger group. A second question asks how errors in these measurements affect an optimal sequential diagnostic policy.

The mathematical ingredients have established antecedents. Pachter and Speyer [20] study reconstruction of trees from subtree weights. Hunter [10] uses spanning trees to optimize bounds on unions from first- and second-order information. Hierarchical representations and their stability are studied by Carlsson and Mémoli [4]. Belief-state dynamic programming follows the finite-horizon partially observed control framework of Smallwood and Sondik [22]. Adaptive submodularity [8] and robust observation selection [13] address related acquisition problems under different assumptions and objectives.

We use these foundations to formulate a precise measurement-to-decision problem. The central result is the exact worst-case recovery radius $(2m - 3)\varepsilon$ for the cost of m sources. Its proof includes a matching lower bound for every estimator using the specified measurements. We then derive decision-value stability, policy excess-loss bounds, and a decision-equivalence characterization. Computable intervals for heterogeneous calibration tolerances additionally give an instance-specific policy certificate and a sufficient condition for a robust first action. Tree reconstruction and the Bellman principle serve as established foundations.

The intended application is diagnosis with reusable setup resources, such as a gateway session or inspection fixture that remains active during one investigation. We make the accounting assumptions explicit and use a synthetic manufacturing example. The numerical study examines calibration error, tree structure, and query limits; it does not estimate performance in a physical factory.

1.1. Related work and scope. Tree identification and hierarchical representation. Buneman [3] gives a metric characterization of trees, while Aho et al. [1] construct trees subject to common-ancestor constraints. Pachter and Speyer [20] show, for $k \geq 3$, that the weights of all k -leaf subtrees identify a positive weighted unrooted tree with n labeled leaves and no degree-two vertices when $n \geq 2k - 1$. Their target is the entire tree. Here a fixed root is included in every acquisition, zero edge lengths are allowed, and the target is a set-cost functional. Our error guarantee does not require separation between branching depths or identification of the physical topology. These differences delimit the claim; the rooted-tree representation itself is established background. Here the measurements are acquisition costs for one or two labeled sources, and the target is the cost of a specified source set. Proposition 2.1 expresses the relevant hierarchy through common-prefix depths; it does not identify a unique physical subdivision of the network. The spanning-tree relation between single linkage and hierarchical clustering studied by Gower and Ross [9] provides context for the overlap repair in Section 3.2. Carlsson and Mémoli [4, Lemma 15 and Proposition 26] establish stability properties of single linkage. Our repair uses

the similarity version of the bottleneck construction; its direct proof also covers raw weights that are not a metric.

Recovery from inaccurate measurements. Optimal recovery asks what can be determined about a target quantity from limited information and a specified model class; Micchelli and Rivlin [16] survey this framework. Donoho [6] connects this viewpoint to minimax statistical estimation of linear functionals under Gaussian noise. Foucart and Liao [7] study inaccurate linear observations in Hilbert spaces, with approximation and observation errors controlled by norm bounds, and characterize optimal regularization choices. The present formulation uses deterministic coordinatewise errors and a nonconvex class of rooted-tree costs. The target is a minimum of finitely many linear forms in the measurement table, rather than a Hilbert-space element under a subspace approximation model. Theorem 3.1 concerns the exact error radius of a set-cost functional under this information model. Its lower bound is obtained from two admissible trees consistent with the same measurements. Thus the result addresses unavoidable uncertainty in acquisition cost without requiring recovery of a particular tree topology.

Sequential testing and partially observed control. Wald [23] develops sequential hypothesis testing, and Chernoff [5] studies the sequential choice of experiments. Naghshvar and Javidi [17] analyze active sequential hypothesis testing through information acquisition, decision penalties, and asymptotically optimal policies. The finite-horizon belief-state formulation used here belongs to the framework of Smallwood and Sondik [22] and the planning treatment of Kaelbling, Littman, and Cassandra [12]. Our model restricts each source to one query and records the queried set because shared activation makes the next acquisition cost depend on earlier choices. The Bellman recursion is a specialization of this established framework.

Submodularity and active feature acquisition. Nemhauser, Wolsey, and Fisher [18] establish approximation guarantees for monotone submodular maximization under a cardinality constraint. Golovin and Krause [8] identify adaptive diminishing-returns conditions supporting greedy acquisition, while Krause et al. [13] study observation selection against multiple possible objectives, including extensions with communication and path costs. These results motivate scalable acquisition methods, but their assumptions concern the information objective as well as its constraints. Submodularity of the shared acquisition cost alone does not establish the required diminishing returns in diagnostic value; the parity example in Section 4.2 makes that distinction explicit.

Active feature acquisition also studies how to choose additional observations for a partially observed instance. Ma et al. [15] combine a partial variational autoencoder with an expected-information-gain acquisition rule, and Li and Oliva [14] use a generative surrogate model to support acquisition-policy learning. These approaches

address representation and policy computation. The present analysis isolates a complementary question: how calibration error in a structured acquisition-cost model propagates to decision values when the observation model is held fixed. The synthetic example therefore illustrates the mathematical guarantees, rather than providing a benchmark against learned acquisition methods.

2. SHARED COSTS ON A ROOTED TREE

2.1. Model and notation. Let $N = \{1, \dots, n\}$ label evidence sources, and let T be a finite tree rooted at a coordinator r . Source i is associated with a vertex and the unique root-to-source path P_i . Each edge e has activation cost $a_e \geq 0$, and source i has service cost $s_i \geq 0$.

Each source is queried at most once during an investigation. An edge is charged when first activated and remains available until the investigation ends.

For $S \subseteq N$, define

$$(2.1) \quad F(S) = \sum_{e \in \bigcup_{i \in S} P_i} a_e + \sum_{i \in S} s_i, \quad F(\emptyset) = 0.$$

This is a nonnegative, monotone weighted-coverage function. Service costs can be absorbed into private pendant edges of lengths s_i , placing each source label at the new endpoint. Thus all costs may be represented by root-path lengths, allowing zero-length edges.

Write

$$(2.2) \quad c_i = F(\{i\}), \quad c_{ij} = F(\{i, j\}), \quad o_{ij} = c_i + c_j - c_{ij}, \quad i \neq j.$$

The quantity o_{ij} is the cost of the common prefix of the two root paths. Service costs cancel from this overlap. Let $\mathcal{T}(S)$ denote the spanning trees of the complete graph on S , when $|S| \geq 2$.

The model describes shared activation or setup expenses. For calibration, each singleton or pair acquisition begins from the same reset condition; service costs are charged once per queried source, and activated resources persist until that acquisition ends. Bounded errors may represent deterministic metering tolerances or externally validated calibration limits. The theory does not estimate those limits from repeated noisy data. A hierarchical gateway or fixture arrangement motivates the rooted-path assumption, which must be justified independently of pair-table compatibility. Expenses that recur per packet, time-dependent latency, congestion, repeated service, and resource deactivation are outside this model.

2.2. Compatibility and marginal costs.

Proposition 2.1 (Rooted-tree compatibility). *A singleton and pair table admits a representation of the form (2.1) if and only if $c_i \geq 0$, the overlaps are symmetric, and*

$$(2.3) \quad 0 \leq o_{ij} \leq \min(c_i, c_j), \quad o_{ij} \geq \min(o_{ik}, o_{jk})$$

for distinct relevant indices. Under these conditions,

$$(2.4) \quad F(A \cup \{j\}) - F(A) = c_j - \max_{i \in A} o_{ij}, \quad j \notin A,$$

where the maximum over the empty set is zero.

Proof. In a rooted tree, intersections of root paths are common prefixes. Their lengths give the bounds in (2.3). If the paths to i and k agree to depth u , and those to j and k agree to depth v , the paths to i and j agree to depth at least $\min(u, v)$. This proves the last inequality.

Conversely, at height $t \geq 0$, group distinct labels satisfying $o_{ij} \geq t$, retaining each label as equivalent to itself. The last inequality in (2.3) makes these groups equivalence classes. As t increases, the classes refine and form a hierarchy.

Realize this hierarchy as a rooted tree whose branching depths are the distinct overlap values, with root depth zero. Place label i at depth c_i . Every nontrivial class containing i occurs at depth at most c_i , so all edge lengths are nonnegative. Zero-length edges accommodate tied depths. The resulting common-prefix depths are exactly o_{ij} , giving the prescribed individual and pair costs.

Finally, the intersection of P_j with the union of previously activated paths is its longest common prefix with those paths. Subtracting this already-paid length from c_j proves (2.4). $\square$

This construction uses the established correspondence between hierarchical similarities and rooted trees [4, 20]. It identifies the set-cost function. A physical subdivision of edges, or a separate decomposition into service and communication expenses, need not be identifiable.

2.3. Reconstruction from pairs.

Proposition 2.2 (Spanning-tree reconstruction). *For every $S \subseteq N$ with $m = |S| \geq 2$,*

$$(2.5) \quad F(S) = \sum_{i \in S} c_i - \max_{M \in \mathcal{T}(S)} \sum_{\{i,j\} \in E(M)} o_{ij}.$$

Equivalently,

$$(2.6) \quad F(S) = \min_{M \in \mathcal{T}(S)} \left[\sum_{\{i,j\} \in E(M)} c_{ij} - \sum_{i \in S} (\deg_M(i) - 1)c_i \right].$$

Proof. Let $W(S)$ be the maximum spanning-tree weight in (2.5). Choose a pair i, j of maximum overlap h in S . Equation (2.4) gives

$$F(S) = F(S \setminus \{j\}) + c_j - h.$$

There is a maximum-weight spanning tree containing edge ij : adding this globally maximum-weight edge to a maximum tree and removing an edge from the resulting cycle cannot decrease its weight.

Moreover, (2.3) implies $o_{ik} \geq o_{jk}$ for every $k \notin \{i, j\}$, since $h \geq o_{jk}$. In a maximum tree containing ij , replace every other edge jk by ik . The result remains a spanning tree, its weight does not decrease, and j becomes a leaf. Its restriction to $S \setminus \{j\}$ must also be maximum. Consequently,

$$W(S) = W(S \setminus \{j\}) + h.$$

Induction, beginning with a singleton of spanning-tree weight zero, proves (2.5). Substituting (2.2) and collecting singleton coefficients proves (2.6). $\square$

The expression has the same spanning-tree structure as Hunter's union bound [10]; rooted-path structure makes it exact. A dense maximum-spanning-tree algorithm evaluates a specified m -source cost in $O(m^2)$ arithmetic operations. All subset costs are represented by $O(n^2)$ measurements, although explicitly listing them requires exponential space.

3. MINIMAX RECOVERY UNDER MEASUREMENT ERROR

Suppose measurements satisfy

$$(3.1) \quad |\tilde{c}_i - c_i| \leq \varepsilon, \quad |\tilde{c}_{ij} - c_{ij}| \leq \varepsilon.$$

The measured table need not satisfy (2.3). Define $H(\emptyset) = 0$, $H(\{i\}) = \tilde{c}_i$, and, for $|S| \geq 2$, define $H(S)$ by the right side of (2.6) using measured values.

3.1. Exact recovery radius.

Theorem 3.1 (Sharp error bound and minimax optimality). *Set*

$$\alpha_0 = 0, \quad \alpha_1 = 1, \quad \alpha_m = 2m - 3 \quad (m \geq 2).$$

Then

$$(3.2) \quad |H(S) - F(S)| \leq \alpha_{|S|}\varepsilon.$$

For each $m \geq 2$, the coefficient $2m - 3$ cannot be reduced by replacing H with any other estimator based only on the individual and pairwise measurements.

Proof. Fix S of size $m \geq 2$. Each spanning-tree expression in (2.6) contains $m - 1$ pair terms with coefficient $+1$. Since every vertex has degree at least one,

$$\sum_{i \in S} |\deg_M(i) - 1| = 2(m - 1) - m = m - 2.$$

Thus each expression changes by at most

$$(m - 1)\varepsilon + (m - 2)\varepsilon = (2m - 3)\varepsilon.$$

If every member of a finite family of real numbers changes by at most d , its minimum changes by at most d . Applying this fact to (2.6) proves (3.2). The empty and singleton cases are immediate.

For the lower bound, consider m leaves sharing a trunk of length $a > 3\varepsilon$, with each pendant edge of length $b > 2\varepsilon$. Let the measured table be that of this central tree:

$$\tilde{c}_i = a + b, \quad \tilde{c}_{ij} = a + 2b.$$

Two possible true trees have trunk and pendant lengths

$$(a_+, b_+) = (a - 3\varepsilon, b + 2\varepsilon),$$

$$(a_-, b_-) = (a + 3\varepsilon, b - 2\varepsilon).$$

All edge lengths are nonnegative. In the first tree, every singleton is lower than its measurement by ε , while every pair is higher by ε . The signs are reversed in the second tree. Both trees satisfy (3.1) for the same observed table.

Their full-set costs are

$$F_+(N) = a + mb + (2m - 3)\varepsilon,$$

$$F_-(N) = a + mb - (2m - 3)\varepsilon.$$

An estimator receives identical input in these two cases. Its largest error must be at least half the distance between the two possible costs, namely $(2m - 3)\varepsilon$. Together with (3.2), this proves minimax optimality. $\square$

Formally, let $\mathcal{C}_m$ be the class of nonnegative rooted-tree cost functions on m labels, and let $D(F)$ be the vector of singleton and pair costs. Then

$$(3.3) \quad \inf_R \sup_{\substack{F \in \mathcal{C}_m, \\ \|z - D(F)\|_\infty \leq \varepsilon}} |R(z) - F(N)| = (2m - 3)\varepsilon, \quad m \geq 2.$$

The infimum ranges over estimators R .

Corollary 3.2 (Unequal singleton and pair tolerances). *If singleton errors are bounded by ε_1 and pair errors by ε_2 , the sharp recovery radius for $m \geq 2$ is*

$$d_m = (m - 2)\varepsilon_1 + (m - 1)\varepsilon_2.$$

Proof. The coefficient count in Theorem 3.1 gives the upper bound. For the lower bound, choose $a > 2\varepsilon_1 + \varepsilon_2$ and $b > \varepsilon_1 + \varepsilon_2$. Starting with a common trunk a and equal pendants b , use true trees with parameters

$$a_\pm = a \mp (2\varepsilon_1 + \varepsilon_2), \quad b_\pm = b \pm (\varepsilon_1 + \varepsilon_2).$$

Their singleton deviations from the central measurements are $\mp\varepsilon_1$, their pair deviations are $\pm\varepsilon_2$, and their full-set costs differ by $2d_m$. The same indistinguishability argument proves the lower bound. $\square$

3.2. A feasible repair. The direct estimate H can be nonmonotone when measurements are incompatible. A physical network representation can instead be obtained by repairing the overlaps.

Define

$$\hat{c}_i = \max(0, \tilde{c}_i),$$

$$(3.4) \quad u_{ij} = \min\{\hat{c}_i, \hat{c}_j, \max(0, \tilde{c}_i + \tilde{c}_j - \tilde{c}_{ij})\}.$$

For distinct i, j , define the maximum-bottleneck closure

$$(3.5) \quad \hat{o}_{ij} = \max_{\gamma: i \rightsquigarrow j} \left\{ \min_{\{u,v\} \in E(\gamma)} u_{uv} \right\},$$

where the maximum is over simple paths in the complete graph. Let $\hat{F}$ be the rooted-tree cost reconstructed from $\hat{c}, \hat{o}$.

Proposition 3.3 (Feasibility and repair error). *The repaired table satisfies (2.3). For every nonempty S ,*

$$(3.6) \quad |\hat{F}(S) - F(S)| \leq (4|S| - 3)\varepsilon.$$

Proof. The closure is symmetric and nonnegative. Every path from i to j starts with an edge bounded by $\hat{c}_i$ and ends with one bounded by $\hat{c}_j$. Hence

$$\hat{o}_{ij} \leq \min(\hat{c}_i, \hat{c}_j).$$

Concatenating paths through k , and removing loops without reducing the bottleneck, proves the final inequality in (2.3).

Singleton clipping has error at most ε . Raw overlap errors are at most 3ε . The clipping in (3.4) preserves this bound because minimum and maximum are nonexpansive in the sup norm.

The maximum-bottleneck operation is also nonexpansive: perturbing every edge value by at most d perturbs each path bottleneck, and then their maximum, by at most d . This operation fixes the true overlap matrix, because repeated application of (2.3) bounds every path bottleneck by the direct true overlap. Therefore,

$$\|\hat{o} - o\|_\infty \leq 3\varepsilon.$$

In (2.5), the singleton sum changes by at most $|S|\varepsilon$, while the maximum spanning-tree weight changes by at most $3(|S| - 1)\varepsilon$. Adding these bounds proves (3.6). $\square$

The closure is a standard hierarchical operation related to single linkage [4, 9]. A max-min version of Floyd's algorithm computes it in $O(n^3)$ time. The constant in (3.6) is a conservative guarantee; its sharpness is not asserted. Repair supplies nonnegative, monotone physical set costs, while direct recovery has the sharper guarantee in (3.2).

3.3. Computable intervals with heterogeneous tolerances. Calibration may provide a different tolerance for each measurement. Write the measured values as z_i, z_{ij} and suppose

$$|z_i - c_i| \leq e_i, \quad |z_{ij} - c_{ij}| \leq e_{ij}, \quad e_i, e_{ij} \geq 0.$$

For a spanning tree M on S , $|S| \geq 2$, put

$$A_M(z) = \sum_{\{i,j\} \in E(M)} z_{ij} - \sum_{i \in S} (\deg_M(i) - 1) z_i,$$

$$d_M = \sum_{\{i,j\} \in E(M)} e_{ij} + \sum_{i \in S} (\deg_M(i) - 1) e_i.$$

Define the preliminary endpoints

$$L_0(S) = \min_{M \in \mathcal{T}(S)} \{A_M(z) - d_M\}, \quad U_0(S) = \min_{M \in \mathcal{T}(S)} \{A_M(z) + d_M\}.$$

Proposition 3.4 (Certified cost intervals). *For $|S| \geq 2$, define*

$$L(S) = \max \left\{ 0, L_0(S), \max_{i \in S} (z_i - e_i), \max_{\{i,j\} \subseteq S} (z_{ij} - e_{ij}) \right\},$$

$$U(S) = \min \left\{ U_0(S), \sum_{i \in S} (z_i + e_i) \right\}.$$

For singletons set $L(\{i\}) = \max\{0, z_i - e_i\}$ and $U(\{i\}) = z_i + e_i$, and set $L(\emptyset) = U(\emptyset) = 0$. Every rooted-tree cost consistent with the measurements satisfies

$$L(S) \leq F(S) \leq U(S).$$

Each endpoint for a specified set of m sources can be evaluated in $O(m^2)$ arithmetic operations. The endpoints need not themselves be rooted-tree cost functions.

Proof. For every M , the coefficient signs give $|A_M(z) - A_M(c)| \leq d_M$. Taking minima and using (2.6) proves $L_0(S) \leq F(S) \leq U_0(S)$. Nonnegativity and monotonicity imply all the additional lower bounds, while $F(S) \leq \sum_{i \in S} c_i$ gives the upper bound. Singleton and empty-set bounds are immediate.

For computation, $A_M(z) - d_M$ is the expression (2.6) with singleton entries $z_i + e_i$ and pair entries $z_{ij} - e_{ij}$. The expression $A_M(z) + d_M$ uses singleton entries $z_i - e_i$ and pair entries $z_{ij} + e_{ij}$. Each minimum therefore requires one spanning-tree calculation, irrespective of compatibility of the shifted table. $\square$

These are computable outer bounds, not a claim of exact pointwise identification limits or heterogeneous minimax optimality. A negative interval width detects inconsistency with the assumptions, but nonempty intervals do not certify the physical rooted-path model. Under a common tolerance ε , $d_M = (2m - 3)\varepsilon$ for every M , so $L_0(S) = H(S) - (2m - 3)\varepsilon$ and $U_0(S) = H(S) + (2m - 3)\varepsilon$.

4. SEQUENTIAL BAYESIAN DECISIONS

4.1. Exact finite-horizon solution. Let the latent state X take values in a finite set $\mathcal{X}$, with prior p . Source i returns an observation in a finite set $\mathcal{Y}_i$, according to a known likelihood $L_i(y \mid x)$. Source observations are conditionally independent given X .

A final decision d belongs to a finite set $\mathcal{D}$, with loss $0 \leq \ell(d, x) \leq 1$. A policy may stop or query an unqueried source, with at most $B \leq n$ queries. If its final queried set is S_τ , its objective is

$$(4.1) \quad J_F(\pi) = \mathbb{E}^\pi[\ell(d_\tau, X) + \lambda F(S_\tau)], \quad \lambda > 0.$$

The query limit is a call budget; monetary cost appears as a penalty. A hard monetary constraint would make the admissible policy class depend on the estimated cost and requires separate treatment.

For posterior p , define

$$(4.2) \quad r(p) = \min_{d \in \mathcal{D}} \sum_x p(x) \ell(d, x),$$

$$q_i(y \mid p) = \sum_x p(x) L_i(y \mid x), \quad T_i(p, y)(x) = \frac{p(x) L_i(y \mid x)}{q_i(y \mid p)}.$$

Outcomes of zero predictive probability are omitted.

Proposition 4.1 (Bellman recursion). *The minimum expected remaining loss and future acquisition cost, after querying A and with b calls remaining, is*

$$(4.3) \quad V_0(p, A) = r(p),$$

and, for $b \geq 1$,

$$(4.4) \quad V_b(p, A) = \min\{r(p), \min_{i \notin A} [\lambda (F(A \cup \{i\}) - F(A)) + \sum_y q_i(y \mid p) V_{b-1}(T_i(p, y), A \cup \{i\})]\}.$$

If no source remains, only the stopping term is used. A deterministic optimal policy exists.

Proof. Conditional independence makes (p, A, b) sufficient for predicting every remaining observation. Stopping incurs Bayes risk $r(p)$. Querying i incurs its marginal cost and the optimal continuation value under the updated posterior.

Backward induction over b proves (4.3)–(4.4). The action and observation sets are finite, so every minimum is attained. Marginal costs telescope to $F(S_\tau)$, giving (4.1) at the initial state. $\square$

This specializes finite-horizon belief-state dynamic programming [12, 22]. With at most q outcomes per source, there are at most

$$\sum_{m=0}^B \binom{n}{m} q^m$$

partial observation assignments from a fixed prior. Compact representation of acquisition costs therefore does not make exact adaptive optimization polynomial.

The same recursion applies to H , using differences $H(A \cup \{i\}) - H(A)$. When raw measurements are incompatible, these differences must be evaluated from the set estimate: formula (2.4) need not hold for noisy overlaps. Negative surrogate increments do not invalidate finite-horizon optimization, although they lack a physical activation-cost interpretation.

4.2. Stability of values and policies. Write $V_F = \min_\pi J_F(\pi)$, $V_H = \min_\pi J_H(\pi)$, and

$$(4.5) \quad \beta_B = \max_{0 \leq m \leq B} \alpha_m = \begin{cases} 0, & B = 0, \\ 1, & B = 1, \\ 2B - 3, & B \geq 2. \end{cases}$$

Theorem 4.2 (Decision stability). *Under (3.1), for the same likelihoods, loss, prior, call budget, and penalty weight,*

$$(4.6) \quad |V_F - V_H| \leq \lambda \beta_B \varepsilon.$$

If π_H minimizes the measured-cost objective, then

$$(4.7) \quad 0 \leq J_F(\pi_H) - V_F \leq 2\lambda \beta_B \varepsilon.$$

For $B \geq 2$, the coefficient in (4.6) is sharp over this decision-model class.

Proof. Fix any history-dependent policy π . Holding this policy fixed, replacing the cost table does not change the observation law or its stopping decisions. Since $|S_\tau| \leq B$, Theorem 3.1 gives

$$\begin{aligned} |J_F(\pi) - J_H(\pi)| &\leq \lambda \mathbb{E}^\pi |F(S_\tau) - H(S_\tau)| \\ &\leq \lambda \beta_B \varepsilon. \end{aligned}$$

Taking minima proves (4.6). If π_F is optimal for F , then

$$\begin{aligned} J_F(\pi_H) &\leq J_H(\pi_H) + \lambda \beta_B \varepsilon \\ &\leq J_H(\pi_F) + \lambda \beta_B \varepsilon \\ &\leq V_F + 2\lambda \beta_B \varepsilon, \end{aligned}$$

which proves (4.7).

For sharpness of (4.6), take B independent fair bits $X_1, \dots, X_B$ as the latent state. Source i reveals X_i deterministically, and the final decision predicts their parity under zero-one loss. Conditional independence holds given the complete latent state. Every proper subset leaves parity uniformly distributed, while the full set determines it.

For a nonnegative monotone cost function K , a policy completing all queries with probability u has objective at least

$$u\lambda K(N) + (1-u)/2.$$

Always stopping or always completing attains the two endpoints. The optimal value is consequently

$$\min\{1/2, \lambda K(N)\}.$$

Use the central tree and the plus perturbation from Theorem 3.1, choosing λ small enough that full acquisition is optimal for both. Their values differ by exactly

$$\lambda(2B-3)\varepsilon.$$

This proves sharpness. □

The factor two in (4.7) is an upper bound; its sharpness is not asserted. Replacing H by the repaired estimate gives corresponding value and excess-loss bounds with $4B-3$, for $B \geq 1$.

The parity example also shows a limitation of one-step policies. If every initial acquisition has positive cost, a policy comparing only immediate Bayes-risk reduction with acquisition cost stops without querying. Its value remains $1/2$, while the optimum is $\lambda F(N)$ for sufficiently small λ . Their ratio is unbounded as λ approaches zero. Thus tree structure alone does not provide a general one-step approximation guarantee; assumptions on information value are also required [8].

The stability question here differs from optimization over uncertain transition laws. Iyengar [11] and Nilim and El Ghaoui [19] develop robust dynamic programming for transition uncertainty. In Theorem 4.2 the likelihoods and admissible policies are common to the true-cost and measured-cost problems; only the acquisition objective changes. Extending the result to uncertain likelihoods would therefore require an additional argument.

4.3. Certificates for a recovered-cost policy. The intervals in Proposition 3.4 can be used directly in a finite decision problem. Let $\mathcal{P}_B$ be the common policy class, and suppose $L(S) \leq F(S) \leq U(S)$ for every $|S| \leq B$, with $L(\emptyset) = U(\emptyset) = 0$. Define J_K by (4.1) for $K = L, U$ and put $V_K = \min_{\pi \in \mathcal{P}_B} J_K(\pi)$. These surrogate set functions may be nonmonotone; their differences can still be used in the finite-horizon recursion.

Proposition 4.3 (Policy and initial-action certificates). *Let π_U minimize J_U . Then*

$$V_L \leq V_F \leq V_U, \quad 0 \leq J_F(\pi_U) - V_F \leq V_U - V_L,$$

and

$$V_U - V_L \leq \lambda \max_{|S| \leq B} \{U(S) - L(S)\}.$$

If $J_U(\pi) \leq V_U + \eta$ for $\eta \geq 0$, the corresponding excess-objective bound is $V_U - V_L + \eta$.

For an admissible initial action a (stopping or a first query), let

$$Q_K(a) = \min_{\pi \in \mathcal{P}_B: \pi \text{ starts with } a} J_K(\pi).$$

If

$$Q_U(a) < \min_{b \neq a} Q_L(b),$$

then a is the unique optimal initial action for every true cost function consistent with the intervals, for the fixed observation model and decision problem.

Proof. For every fixed policy, $J_L(\pi) \leq J_F(\pi) \leq J_U(\pi)$. Minimizing gives the value ordering, and applying it to π_U gives the regret bound. If π_L minimizes J_L , then

$$V_U - V_L \leq J_U(\pi_L) - J_L(\pi_L) \leq \lambda \max_{|S| \leq B} \{U(S) - L(S)\}.$$

The approximate-policy assertion follows by adding η . The same objective ordering within each class of policies having a specified first action yields $Q_L(b) \leq$

$Q_F(b) \leq Q_U(b)$. The strict separation condition therefore gives $Q_F(a) < Q_F(b)$ for every alternative b . $\square$

This is a sufficient certificate: its failure does not imply that an action is unstable. For a common measurement tolerance, the interval-width bound is at most $2\lambda\beta_B\varepsilon$. Solving the lower and upper decision problems can yield a smaller, instance-specific certificate. The upper-cost policy is not asserted to have the least true regret, nor to solve a minimax problem over mutually compatible cost tables. The argument is an objective perturbation principle, applied here to the computable intervals of the rooted-tree model.

5. DECISION EQUIVALENCE AND THE BOUNDARY OF THE MODEL

5.1. Which network information is decision-relevant?

Theorem 5.1 (Universal decision equivalence). *Let F and G be rooted-tree cost functions on the same $n \geq 2$ labels. Fix $2 \leq B \leq n$. The following statements are equivalent:*

1. *F and G agree on every singleton and pair.*
2. *They agree on every subset of sources.*
3. *They have equal optimal values for every finite decision experiment described in Section 4 and every $\lambda > 0$, under the same call limit B .*

Proof. Proposition 2.2 gives statement 1 implies statement 2. Identical set costs give identical objectives for every policy, proving statement 2 implies statement 3.

Suppose statement 1 fails on a set S of size one or two. Let each source in S reveal one independent fair bit, let every other source return a constant, and let the final decision predict the parity of the bits in S .

By the argument in Theorem 4.2 and monotonicity of acquisition costs, the optimal value under a cost function K is

$$\min\{1/2, \lambda K(S)\}.$$

Choose $\lambda > 0$ small enough that acquiring S is optimal under both F and G . Since $F(S) \neq G(S)$, their optimal values differ, contradicting statement 3. $\square$

For $B = 1$, only singleton costs matter. The quantifier over all positive λ is essential: a sufficiently high fixed penalty can make immediate stopping optimal in both networks and conceal cost differences.

Blackwell's comparison of experiments [2] provides a classical decision-theoretic use of universal quantification over decision problems. The objects compared here are acquisition-cost functions: the observation experiment is the same on both sides

of each comparison. Theorem 5.1 characterizes when cost differences can be detected by some such experiment; it is not a characterization of the informational equivalence of observation channels.

5.2. Failure for general shared resources. Rooted-path structure cannot be inferred solely from a compatible pair table. Consider three sources.

In system A, there are three resources e_{12}, e_{13}, e_{23} , each of cost one. Source 1 uses e_{12}, e_{13} ; source 2 uses e_{12}, e_{23} ; source 3 uses e_{13}, e_{23} .

In system B, every source uses a common resource of cost one and its own private resource of cost one.

Both systems have singleton cost two and pair cost three. Their triple costs differ:

$$(5.1) \quad F_A(\{1, 2, 3\}) = 3, \quad F_B(\{1, 2, 3\}) = 4.$$

System B is a rooted-tree model; system A is a general coverage system. Individual and pairwise measurements therefore cannot identify all group costs over the broader class. Even a table satisfying Proposition 2.1 does not certify that the physical system has rooted-path structure.

6. SYNTHETIC MANUFACTURING ILLUSTRATION

6.1. Fully specified instance. Consider six evidence sources: telemetry, quality, maintenance, materials, scheduling, and assembly. Their labels describe diagnostic roles. No physical factory data or language-model outputs are used.

The rooted routing tree has the following activation edges.

Edge	Activation cost
Coordinator $\rightarrow$ Telemetry	2.0
Telemetry $\rightarrow$ Quality	1.6
Telemetry $\rightarrow$ Maintenance	1.4
Coordinator $\rightarrow$ Materials	2.6
Coordinator $\rightarrow$ Scheduling	2.0
Scheduling $\rightarrow$ Assembly	1.8

An activation expense is paid once per investigation.

The three pairs within {telemetry, quality, maintenance}, and the scheduling–assembly pair, have overlap 2.0. Every other overlap is zero. These values and Table 1 specify every pair cost through (2.2).

For example, quality and maintenance together cost

$$4.7 + 4.3 - 2.0 = 7.0.$$

TABLE 1. Synthetic source parameters.

Source	Service cost	Singleton cost	Accuracy
Telemetry	1.0	3.0	0.75
Quality	1.1	4.7	0.80
Maintenance	0.9	4.3	0.78
Materials	0.8	3.4	0.65
Scheduling	0.9	2.9	0.70
Assembly	1.3	5.1	0.88

Adding telemetry costs $3.0 - 2.0 = 1.0$, giving a three-source total of 8.0. Acquiring all six sources costs 17.4; the sum of standalone costs is 23.4. This difference follows from the assumed accounting rule and is not an empirical manufacturing saving.

Let $X \in \{0, 1\}$ have prior probability $1/2$. Each source is a binary symmetric channel with the accuracy in Table 1, independently conditional on X . The final loss is diagnostic misclassification, and at most four sources may be queried.

We compare the Bellman-optimal policy with a myopic policy minimizing, at each history,

$$(6.1) \quad \min\{r(p), \min_{i \notin A} [\lambda (F(A \cup \{i\}) - F(A)) + \sum_y q_i(y | p) r(T_i(p, y))]\}.$$

The myopic policy repeats this rule after each observation. Ties favor stopping, then the earliest source in Table 1.

TABLE 2. Exact policy expectations for the synthetic instance.

λ	Policy	Error probability	Expected cost	Expected calls	Objective
0.005	Optimal	0.0792	7.83012	2.4716	0.1183506
0.005	Myopic	0.1200	5.10000	1.0000	0.1455000
0.010	Optimal	0.0828	7.39230	2.4770	0.1567230
0.010	Myopic	0.1200	5.10000	1.0000	0.1710000
0.020	Optimal	0.1200	5.10000	1.0000	0.2220000
0.020	Myopic	0.1200	5.10000	1.0000	0.2220000
0.040	Optimal	0.1200	5.10000	1.0000	0.3240000
0.040	Myopic	0.1200	5.10000	1.0000	0.3240000

6.2. Exact results. At the two smaller penalty weights, the optimal policy first queries scheduling, whereas the myopic policy queries assembly and stops. Although assembly has greater assumed standalone accuracy, planning ahead accounts for

shared activation and subsequent evidence. At the larger penalty weights, both policies query assembly and stop. These results establish behavior only for the specified finite instance.

For calibration tolerance $\varepsilon = 0.05$, the four-call limit gives $\beta_4 = 5$. At $\lambda = 0.01$, Theorem 4.2 bounds the optimal-value error by 0.0025 and the excess objective of the measured-cost policy by 0.005. These are worst-case guarantees under the calibration assumption, rather than observed average errors.

The supplied Python implementation reproduces Table 2 with exact rational arithmetic using `Fraction`. The Bellman calculation considered 473 partial observation assignments, equal to

$$\sum_{m=0}^4 \binom{6}{m} 2^m.$$

For each policy and penalty, its terminal decision, cost, and query count were independently evaluated by replaying all

$$2 \times 2^6 = 128$$

latent-state/full-observation combinations with their exact probabilities. Across the eight policy cases, all 1,024 replay outcomes agreed exactly with the expectations computed by backward induction. Table 2 reports exact finite-model expectations, rather than Monte Carlo estimates.

6.3. Algebraic implementation checks. The supplied implementation generates 100 rooted trees with six labeled source vertices. For each vertex $i = 1, \dots, 6$, it draws a parent from $0, \dots, i-1$, then an edge weight from $\{0.1, 0.2, \dots, 3.0\}$, then a service cost from $\{0.1, 0.2, \dots, 1.0\}$. Vertex 0 is the root. A 32-bit linear congruential generator uses seed 20260911, multiplier 1664525, and increment 1013904223. The code specifies the complete draw order, including subsequent perturbations.

For all 64 subsets of each tree, exact path-union costs agree with Proposition 2.2: 6,400 equalities. All 19,200 source-addition marginals agree with (2.4). With measurement perturbations in $\{-1/20, 0, 1/20\}$, all 6,400 direct and 6,400 repair bounds hold; every repaired table satisfies (2.3). The code also checks 6,400 interval inclusions with uniform tolerances and 6,400 with heterogeneous tolerances from $\{0.01, 0.02, \dots, 0.05\}$. The two signs of the sharpness construction are checked for $m = 2, \dots, 8$, under both equal and unequal singleton/pair tolerances, giving 28 exact checks. These finite calculations verify the implementation; the universal claims follow from the proofs.

6.4. Sensitivity to calibration and tree structure. We retain the service costs and binary observation channels of Table 1 and compare four cost structures. The original is the routing tree in Section 6.1. A star has a common trunk of cost 2 and private edges of costs $(1, 1.6, 1.4, 1.8, 1.2, 2)$. A balanced hierarchy has a common trunk of cost 1, three branches of costs $(0.8, 1.2, 0.9)$ serving consecutive pairs of sources, and private edges of costs $(0.6, 0.9, 0.7, 0.8, 0.5, 1)$. A chain places the six sources at successive vertices, with edge costs $(1.2, 0.8, 1, 0.7, 0.9, 1.1)$. Service costs are added in every structure. These choices vary shared activation without changing the observation experiment.

The call limits are two, four, and six queries. The four penalty weights are 0.005, 0.01, 0.02, and 0.04. Calibration tolerances are 0.01, 0.05, 0.2, and 0.5. For each combination we generate ten measurement perturbations from 201 equally spaced values in $[-\varepsilon, \varepsilon]$, using the same generator with seed 20260913. Thus there are $4 \times 3 \times 4 \times 4 \times 10 = 1,920$ cases. These are finite pseudorandom sensitivity cases, not independent experimental samples from a factory population. The supplied code and per-case CSV specify the generation order and all results.

For each case, complete enumeration of partial observation assignments computes optimal policies under $F, H, \widehat{F}, L, U$. Each recovered-cost policy is then evaluated under the true cost F . These calculations use double precision, with 10^{-10} tolerance for bound checks; they do not simulate observation trajectories. The direct-policy regret is $J_F(\pi_H) - V_F$, and the upper-cost policy has certificate $V_U - V_L$. Initial-action certificates use a strict numerical separation exceeding 10^{-12} .

TABLE 3. Calibration sensitivity, aggregated over 480 cases per row.

ε	Mean direct regret	Max. direct regret	Mean $V_U - V_L$	Certified first actions
0.01	8.32×10^{-7}	8.44×10^{-5}	4.94×10^{-4}	58.3%
0.05	1.02×10^{-5}	1.40×10^{-3}	2.48×10^{-3}	57.7%
0.2	1.30×10^{-4}	7.81×10^{-3}	9.93×10^{-3}	47.7%
0.5	9.61×10^{-4}	2.41×10^{-2}	2.42×10^{-2}	21.9%

All checked value, policy, repair, and interval bounds hold. Direct-policy regret exceeds 10^{-10} in 328 cases. The initial-action criterion certifies 891 of the 1,920 cases, and each certified action agrees with the true-cost optimum. Certification becomes less frequent as the tolerance increases. The certificate column concerns π_U , whereas the first two regret columns concern π_H ; it is not a bound on π_H in a particular instance.

The feasible repair and the upper-cost policy do not uniformly dominate the direct estimate. Their mean regrets at $\varepsilon = 0.2$ are approximately 1.02×10^{-4} and $3.02 \times$

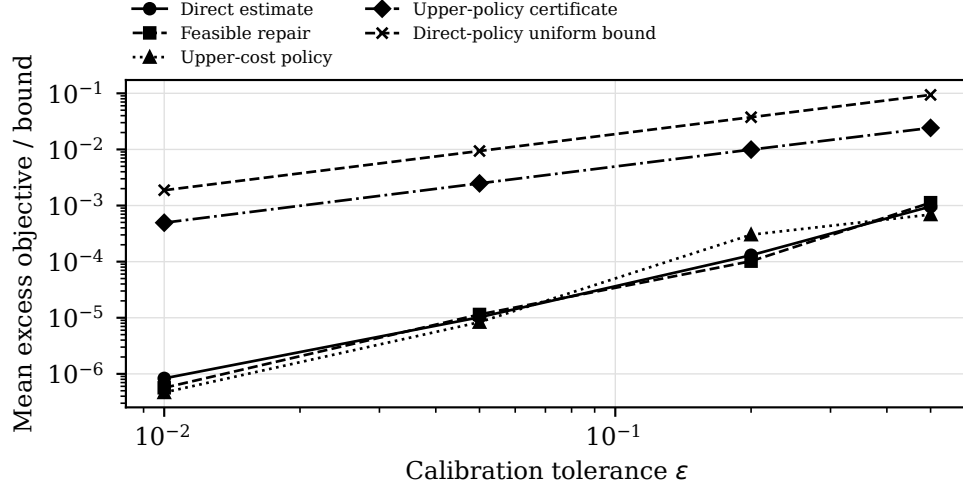


FIGURE 1. Mean excess objectives and bounds over 480 cases at each tolerance. The direct-policy uniform bound is $2\lambda\beta_B\varepsilon$; the upper-policy certificate is $V_U - V_L$. Lines summarize this synthetic grid only.

10^{-4} , respectively, compared with 1.30×10^{-4} for direct recovery. At $\varepsilon = 0.5$ they are 1.12×10^{-3} and 6.91×10^{-4} , compared with 9.61×10^{-4} . Repair provides a physical representation; interval optimization provides a certificate. Neither property alone guarantees smaller true regret. The worst-case bounds remain substantially larger than mean realized regret on this grid, as expected for guarantees uniform over the model class.

7. DISCUSSION AND CONCLUDING REMARKS

The mathematical problem separates into identification, estimation, and decision making. Under the rooted-tree assumption, pairwise overlap data determine every acquisition cost. Under bounded measurement error, the cost of an m -source group has minimax recovery radius $(2m - 3)\varepsilon$. This radius quantifies unavoidable amplification of measurement uncertainty, including cases in which the measured table itself is compatible with a tree.

For sequential decisions, uniform cost recovery transfers directly to optimal-value and policy excess-loss guarantees. Heterogeneous measurement tolerances can instead be propagated through computable set-cost intervals, giving an instance-specific certificate and a sufficient condition for a robust initial action. The transfer does not depend on a particular diagnostic classifier, but it requires an unchanged

observation model and a common call-constrained policy class. Likelihood misspecification, time-varying costs, repeated queries, resource deactivation, and hard monetary budgets require separate analysis.

The application illustrates how a coordinator can use the theory to choose sources and determine when to stop. It does not establish industrial validity or the performance of an implemented large-language-model system. Exact adaptive optimization remains exponential in the number of sources; scalable approximations need additional structure.

The counterexample in Section 5 identifies a substantive boundary: pair measurements suffice within the rooted-path class but do not suffice for general shared resources. Extensions to partially observed routing structures, higher-order calibration measurements, and joint uncertainty in costs and likelihoods provide concrete directions for further research.

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